

***Brock Biology of Microorganisms, 15e (Madigan et al.)***  
**Chapter 4 Molecular Information Flow and Protein Processing**

**4.1 Multiple Choice Questions**

- 1) The functional unit of genetic information is the
- A) nucleotide.
  - B) gene.
  - C) chromosome.
  - D) protein.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding  
Chapter Section: 4.1

- 2) Which of the following statements is TRUE concerning mRNA?
- A) mRNA has a very short half-life.
  - B) mRNA has complex secondary structure.
  - C) mRNA is catalytic.
  - D) mRNA is the product of translation.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding  
Chapter Section: 4.6

- 3) DNA-binding proteins interact predominantly within which portion of a double-stranded DNA helix?
- A) major groove
  - B) minor groove
  - C) 3' end
  - D) supercoil

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding  
Chapter Section: 4.1

- 4) AT-rich DNA will denature/melt
- A) at a higher temperature than GC-rich DNA.
  - B) at a lower temperature than GC-rich DNA.
  - C) usually at the same temperature as GC-rich DNA, with some minor variations.
  - D) in accordance with the animal or plant from which it was taken.

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating  
Chapter Section: 4.1

- 5) Supercoiling is important for DNA structure, because
- A) it holds together the antiparallel strands of DNA in the double helix.
  - B) it provides energy for transcription.
  - C) it condenses the DNA so that it can fit inside the cell.
  - D) it prevents RNA from pairing with DNA in the double helix.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.1

6) Many pharmaceutical drugs specifically inhibit transcription in *Bacteria* but not *Archaea* or *Eukarya*. Why would drugs that inhibit transcription only affect *Bacteria* and not *Archaea* even though they are both prokaryotes?

- A) *Archaea* and *Eukarya* have very similar ribosomes that are different than bacterial ribosomes.
- B) *Bacteria* lack a nucleus.
- C) *Archaea* lack operons.
- D) *Archaea* and *Eukarya* have very similar RNA polymerases that are different than bacterial RNA polymerases.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 4.6

7) How are plasmids different than chromosomes?

- A) Plasmids are always small, linear pieces of DNA.
- B) Plasmids are composed of single-stranded DNA.
- C) Plasmids contain genes that are NOT essential for cellular growth and replication.
- D) Plasmids carry unimportant genes that are of little significance for the ecology and metabolism of an organism.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.2

8) Genes that are encoded for polymerases, gyrases, ribosomal proteins, and other proteins essential to replication, transcription, and translation are present on

- A) chromosomes.
- B) plasmids.
- C) chromosomes and plasmids.
- D) neither chromosomes nor plasmids.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.2

9) The precursor of each new nucleotide in a strand of DNA is a

- A) deoxynucleoside 5'-diphosphate.
- B) deoxynucleoside 3'-diphosphate.
- C) deoxynucleoside 5'-triphosphate.
- D) deoxynucleoside 3'-triphosphate.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

10) DNA replication always proceeds in only one direction because the \_\_\_\_\_ of the incoming nucleotide is attached to the free \_\_\_\_\_ of the growing DNA strand.

- A) 5'-phosphate / 3'-hydroxyl
- B) 3'-phosphate / 5'-hydroxyl
- C) 5'-deoxyribose / 3'-base
- D) 3'-base / 5'-deoxyribose

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

11) Which of the following is formed on the lagging strand during DNA synthesis?

- A) DNA secondary structures
- B) Okazaki fragments
- C) RNA polymerase
- D) replisomes

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

12) The template for RNA polymerase is \_\_\_\_\_, and the new RNA molecule is \_\_\_\_\_ to the template.

- A) an independent RNA segment / parallel and identical
- B) DNA / antiparallel and complementary
- C) an independent RNA segment / antiparallel and complementary
- D) DNA / parallel and identical

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.5

13) In the process of transcription, promoters are specific sequences of \_\_\_\_\_ that are recognized by \_\_\_\_\_.

- A) DNA / DNA polymerase
- B) RNA / DNA polymerase
- C) DNA / sigma factors
- D) RNA / ribosomes

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.5

14) An example of correct nucleotide pairing is

- A) T and U.
- B) G and U.
- C) A and T.
- D) C and U.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.5

15) Stop codons are also called \_\_\_\_\_ codons.

- A) nonsense
- B) release factor
- C) degeneracy
- D) conversion

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.9

16) Transfer RNA (tRNA) molecules

- A) function to transfer ribonucleotides to RNA polymerase during transcription.
- B) function to transfer the correct amino acids to the ribosome during translation.
- C) contain codons that bind to ribosomes during translation.
- D) are only present in the nucleus or eukaryotes.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.8

17) tRNA is released from the ribosome at the \_\_\_\_\_ site.

- A) P
- B) A
- C) R
- D) E

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.9

18) Which statement is TRUE regarding protein synthesis?

- A) Ribosomal proteins catalyze peptide bond formation in the growing polypeptide chain.
- B) The 23S rRNA catalyzes peptide bond formation in the growing polypeptide chain.
- C) Transfer RNAs catalyze peptide bond formation in the growing polypeptide chain.
- D) Messenger RNA catalyzes peptide bond formation in the growing polypeptide chain.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.9

19) The Tat system is involved in

- A) protein synthesis.
- B) transcriptional initiation.
- C) protein folding.
- D) protein secretion.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.12

20) In all cells, genes are composed of

- A) nucleic acids.
- B) mRNA.
- C) proteins.
- D) chaperones.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.1

21) Which of the following is an example of one codon?

- A) CATT
- B) GCCATT
- C) CAG
- D) CCGUAA

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.9

22) In all cells a gene encodes for

- A) a protein (via mRNA).
- B) a tRNA.
- C) an rRNA.
- D) a protein, tRNA, or rRNA depending on the specific gene.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.9

23) Which of the following is NOT correct regarding DNA and RNA synthesis?

- A) The overall direction of chain growth is from the 5' to 3' end.
- B) Both processes require an RNA primer to begin.
- C) The template strand is antiparallel to the newly synthesized strand.
- D) DNA is the template for both DNA and RNA synthesis.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

24) Termination of RNA synthesis is ultimately determined by

- A) exhaustion of RNA polymerase activity.
- B) special protein factors.
- C) terminases.
- D) specific nucleotide sequences on the template strand.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.5

25) GTP provides energy for

- A) transcription.
- B) translation.
- C) DNA replication.
- D) protein folding.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.10

26) Transcription of chaperones is greatly accelerated when a cell is stressed by

- A) excessive osmotic pressure.
- B) extremes in pH value.
- C) lack of oxygen.
- D) excessive heat.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.11

27) The flow of biological information begins with

- A) DNA replication.
- B) RNA transcription.
- C) mRNA translation.
- D) transcriptional regulation.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.1

28) DNA participates in protein synthesis through

- A) cyclic messengers.
- B) direct pairing with amino acids.
- C) an RNA intermediate.
- D) protein folding.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.1

29) The two strands of the DNA double helix are held together by

- A) 5' to 3' attraction.
- B) hydrogen bonds between nucleotide bases.
- C) codons.
- D) peptide bonds between nucleotide bases.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.1

30) In *Bacteria*, a chromosome can be distinguished from a plasmid, because a chromosome is a genetic element that

- A) is circular.
- B) is linear.
- C) encodes for essential functional genes.
- D) replicates via a bidirectional fork.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.2

31) Transposable elements are

- A) segments of DNA that move from one site to another.
- B) transcribed genes.
- C) segments of RNA that are involved in transposing DNA into proteins.
- D) proteins that aid in the secretion of enzymes out of the cell.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.2

32) In complementary base pairing of DNA, adenine pairs with \_\_\_\_\_ (or \_\_\_\_\_ in RNA) and cytosine always pairs with \_\_\_\_\_.

- A) guanine / uracil / thymine
- B) uracil / thymine / guanine
- C) thymine / guanine / uracil
- D) thymine / uracil / guanine

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.1

33) The function of the DNA polymerase is to catalyze

- A) the addition of deoxynucleotides.
- B) the formation of RNA primers.
- C) the addition of ribonucleotides.
- D) hydrogen bonding between complementary base pairs.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

34) DNA replication is started with a(n) \_\_\_\_\_, which, in most cases, *in vivo* is a short stretch of \_\_\_\_\_.

- A) promoter / DNA
- B) mRNA / RNA
- C) primer / RNA
- D) ribosome-binding sequence / DNA

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

35) The function of RNA polymerase is to

- A) catalyze the formation of phosphodiester bonds between deoxyribonucleotids.
- B) catalyze the formation of phosphodiester bonds between ribonucleotides.
- C) cleave mRNA to remove introns.
- D) activate tRNAs.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.5

36) Polycistronic transcription units are common in

- A) *Archaea*.
- B) *Bacteria*.
- C) *Eukarya*.
- D) both *Archaea* and *Bacteria*.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.5

37) An operon is a useful genetic element, because it

- A) encourages the binding of RNA polymerase.
- B) allows coordinated expression of multiple related genes in prokaryotes.
- C) translates DNA sequence into amino acid sequence.
- D) encourages the binding of ribosomes in the correct location.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 4.5

38) Plasmids often encode for proteins

- A) involved in translation.
- B) required for cellular growth.
- C) that confer resistance to antibiotics.
- D) involved in DNA replication.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.2



39) The codon on the \_\_\_\_\_ matches with the anticodon on the \_\_\_\_\_ to direct the addition of the correct amino acid to the growing polypeptide chain.

- A) mRNA / tRNA
- B) tRNA / mRNA
- C) DNA / mRNA
- D) tRNA / rRNA

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.8

40) The structure and function of a protein are determined by its \_\_\_\_\_ sequence.

- A) nucleotide
- B) amino acid
- C) ribonucleotide
- D) translocation

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.9

41) You experimentally change the DNA sequence directly upstream of a start codon of an operon in *E. coli* to investigate the function of this region of DNA. Analysis reveals that after the change the same amount of mRNA is made from the operon, but there are very few proteins made from the operon. What is the most likely function of the DNA sequence that you changed?

- A) The DNA sequence likely functions as a ribosome-binding site.
- B) The DNA sequence likely functions as a promoter.
- C) The DNA sequence likely functions as a termination sequence.
- D) The DNA sequence likely functions as in transcriptional regulation.

Answer: A

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 4.10

42) In \_\_\_\_\_ several ribosomes can simultaneously translate a single mRNA molecule in a complex called a(n) \_\_\_\_\_.

- A) prokaryotes / polysome
- B) eukaryotes / polysome
- C) prokaryotes / initiation complex
- D) eukaryotes / splicing complex

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.10

43) You are studying a protein in *Salmonella typhimurium* that you believe is a toxin. Whenever you attempt to purify the protein from lysed cell cultures, you get two forms of the protein. One form is smaller than the other and is missing 15 amino acids from the N-terminus compared to the larger form. This leads you to hypothesize that

- A) there are two termination sites in the mRNA.
- B) the protein requires chaperonins to fold properly.
- C) the protein is secreted and folds outside of the cell.
- D) you need to re-do the experiment because there should only be one form.

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 4.11

44) Transcription in eukaryotes occurs in the

- A) RNA polymerase.
- B) endoplasmic reticulum.
- C) cytoplasm.
- D) nucleus.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.6

45) During DNA replication Okazaki fragments are linked together by \_\_\_\_\_, an enzyme that creates phosphodiester bonds between nicked fragments of DNA.

- A) exopolymerase
- B) DNA gyrase
- C) topoisomerase
- D) DNA ligase

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

46) A triplet of bases on an mRNA molecule is known as a(n)

- A) amino acid.
- B) anticodon.
- C) codon.
- D) ribosome-binding sequence.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.9

47) Each adenine-thymine base pair has \_\_\_\_\_ hydrogen bonds, while each guanine-cytosine base pair has \_\_\_\_\_ hydrogen bond(s).

- A) two / one
- B) two / three
- C) four / three
- D) three / two

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.1

48) In DNA replication there are leading and lagging strands, because

- A) DNA replication is conservative and a completely new DNA molecule must be made.
- B) DNA replication is semiconservative and each strand is copied simultaneously in opposite directions.
- C) the strands of DNA are parallel and are copied in the same direction simultaneously.
- D) one strand of DNA is copied faster than the other.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

#### 4.2 True/False Questions

1) Most prokaryotic genomes are double-stranded circular DNA.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.2

2) Inverted repeats are common features of transcription termination sequences in all three domains of life.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.5

3) In nature, the predominant form of DNA is supercoiled in a negative direction.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.1

4) Genes found on plasmids do NOT impact metabolism or cellular structures.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.2

5) A plasmid is all the genetic material present in a virus.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.2

6) DNA replication is bidirectional in prokaryotes with circular chromosomes.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.4

7) RNA acts at both the genetic and the functional levels.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.1

8) RNA is incapable of forming secondary structure.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.5

9) Aminoacyl-tRNA synthetase catalyzes the reaction between the appropriate amino acid and ATP to form an activated amino acid: amino acid + ATP  $\leftrightarrow$  aminoacyl-AMP + P-P.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.8

10) The formation of new DNA does NOT require energy.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

11) rRNA has an enzymatic role in all stages of protein synthesis.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.10

12) Proteins known as chaperones are found only in *Bacteria*.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.11

13) Throughout the living world, the genetic code is generally universal; however, there are slight variations.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.9

14) DNA replication involves the synthesis of an RNA primer on one strand of the DNA.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

### 4.3 Essay Questions

1) You isolate a piece of DNA from a microorganism you cultivated from your teeth. The piece of DNA is 94 kbp in size and is circular. You sequence it and discover that it contains genes for capsule formation, pili, and antibiotic resistance, as well as an origin of replication. What is this piece of DNA and how is it related to the other genetic elements found in prokaryotic cells?

Answer: The piece of DNA is most likely a plasmid. Plasmids are usually less than 1 Mbp in size and are circular. Chromosomes are much larger and can be circular or linear, although they are usually circular in prokaryotes. Plasmids usually carry genes for antibiotic resistance and other special metabolic or structural features that impact cell ecology and virulence, but are not absolutely essential for cellular growth and replication. Plasmids can also replicate independently from the chromosome, although they use the host cell's DNA polymerase and other enzymes for replication. Plasmids are different than viruses because they are always made of double-stranded DNA, are circular, and contain origins of replication.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 4.2

2) The following is the sequence of bases in the sense strand of a DNA segment and contains the beginning of a gene.

**DNA 3' A T A T T A C C A G G C A T G G A C C C C C G G G 5'**

Based on this sequence, write the sequence of the anti-sense DNA strand and the mRNA. Label the 5' and 3' ends in your predicted sequences. The start codon in this organism is AUG. Indicate where the start codon is in your sequence. Why is the start codon important? Why does there have to be a specific start codon?

Answer: Anti-Sense: 5' T A T T T A G G T C C G T A C C T G G G G G C C C 3' mRNA: 5' U A U A **AUG** G U C C G U A C C U G G G G G C C C 3'

The start codon is bolded and underlined above and should only be indicated in the mRNA molecule. A start codon is important because it tells the ribosome where to start reading the codons so that the gene is read in the correct frame. Without a start signal (codon) to indicate which nucleotide to start reading the three letter codons from, the frame could be shifted. This would completely change the amino acids that are read and could result in "nonsense," or a non-functional protein.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 4.9

3) What is the basic flow of genetic information in all cellular life? Include in your answer a diagram that illustrates the relationships between the basic components and steps in the flow of genetic information.

Answer: Answers and diagrams should demonstrate understanding of the central dogma of molecular biology (DNA RNA Protein) and the basic enzymes/molecules involved in each step (RNA polymerase, ribosomal proteins, rRNA, and tRNA). Answers could also explain that codons direct the order of amino acids in a protein, but the question does not explicitly ask for that information.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.1

4) Explain the difference between transcription and translation and how the processes differ in bacteria and eukaryotes.

Answer: Transcription is the process in which mRNA transcripts are synthesized from DNA, whereas translation uses RNA as a template to synthesize peptides (or proteins). Transcription and translation occur simultaneously in the cytoplasm in bacteria, but in eukaryotes transcription occurs in the nucleus and translation takes place in the cytoplasm and endoplasmic reticulum. Transcripts are not significantly modified in bacteria before translation begins, but significant modification of mRNA occurs in eukaryotes before translation can begin. Eukaryotic mRNA contains introns that must be removed before translation. Capping and poly-A tail addition must also occur for the mRNA to be exported from the nucleus and translated in eukaryotes.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.6

5) Explain the concept of semiconservative replication and how simultaneous copying of both strands of DNA is accomplished in prokaryotic cells.

Answer: During replication, only one of the two strands is used as a template (called the parental strand) to form a complementary strand called the progeny (or daughter) strand. Therefore, because only one of the two parental strands is needed for replication, it is considered partially (or semi-) conservative. Simultaneous copying of both strands of DNA is accomplished by opening the DNA double helix and copying the DNA in both directions, forming a replication bubble. One of the DNA strands (the leading strand) is continuously copied by DNA polymerase in the 5' to 3' direction. The other strand of DNA (the lagging 3' to 5' strand) must be copied in short fragments, because the direction of synthesis is the opposite of the movement of the replication fork. This occurs because DNA can only be synthesized in the 5' to 3' direction. Synthesis of DNA on the lagging strand must be repeatedly primed and extended, then the RNA primer is removed and replaced with DNA. The final step, which is performed by DNA ligase, is to join all of the DNA fragments together in a completed continuous daughter strand. (The level of detail in this question should commensurate with the level of detail discussed in class with the students.)

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

6) Discuss how the initiation of DNA synthesis occurs in bacteria using the terms *origin of replication*, *replication fork*, and *theta structures*.

Answer: The initiation protein DnaA binds to the origin of replication (*oriC* gene) where DnaB and DnaC proteins help unwind the DNA. This unwound and accessible region of DNA, called the replication fork, is where DNA synthesis occurs. Theta structures are used to describe the appearance of the DNA replication during bidirectional circular DNA replication.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

7) Explain the function of the helicases and why they are necessary.

Answer: Helicase enzymes require energy from ATP to unwind double-stranded DNA into single-stranded (ss) DNA. Its function is necessary during the initiation of DNA synthesis, because the polymerase can operate only on ssDNA as the template.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

8) Some essential genes and DNA sequences in cells do NOT encode for proteins but are still essential for cellular growth and replication. Give two examples of a gene or sequence of which this is TRUE and explain why it is essential for growth or replication.

Answer: Answers to this question could use many different examples such as rRNA and tRNA encoding genes. These genes are NOT translated into protein but are transcribed. The RNA that is made after transcription is the final end product that is critical for translation. Without rRNA or tRNA genes a cell would not be able to make new proteins or grow. Other essential DNA sequences are promoter sequences, ribosome binding sites, and termination sequences.

Promoters and ribosome binding sites are never translated, but they are critical to the expression of all genes in a cell.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 4.5

9) Explain the role of sigma factors in RNA synthesis in *Bacteria*.

Answer: A sigma factor is one of five subunits within an RNA polymerase involved exclusively in transcription of DNA to RNA. It identifies and binds to a promoter (initiation) site, notably the highly conservative -10 and -35 sites, which creates an RNA polymerase-DNA template complex that facilitates transcription initiation. A sigma factor then dissociates from the template after a short stretch of RNA is synthesized. They are also subject to regulation by anti-sigma factor proteins that temporarily inactivate a sigma factor and thus halt RNA synthesis.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.5

10) Explain the process of RNA transcription using the terms *upstream*, *Pribnow box*, and *consensus sequence*.

Answer: The RNA polymerase holoenzyme initiates transcription without the need for a primer. The sigma factor component of RNA polymerase easily dissociates from the holoenzyme to first recognize the promoter region by specifically binding to the Pribnow box and consensus sequence, which are both upstream of the transcriptional start site.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.5

11) Explain the difference between an intrinsic terminator and a Rho-dependent termination site.  
Answer: Intrinsic terminators occur at the nucleotide (DNA) level and require no additional protein involvement. They can develop into secondary structures such as loops, which freely form through base pairing with itself to ultimately halt transcription of RNA. Rho-dependent termination also stops RNA transcription but requires the protein Rho, which interacts with the DNA template-RNA polymerase complex to dissociate the complex at specific nucleotide sequences.

Bloom's Taxonomy: 1-2: Remembering/Understanding  
Chapter Section: 4.5

12) How is an open reading frame (ORF) identified and used to determine the sequence of amino acids in the gene it encodes for?

Answer: An ORF is often predicted by identifying a start codon, followed by a nucleotide sequence, and a terminating nonsense codon in the same reading frame. From the ORF sequence, the amino acid sequence can be predicted by translating each individual codon, which aids in predicting biochemistry and function of the predicted protein.

Bloom's Taxonomy: 3-4: Applying/Analyzing  
Chapter Section: 4.9

13) Is the following statement TRUE or FALSE? Explain why you think so. "Proteins are the only biomolecules capable of catalyzing bond formation."

Answer: A theme to this answer should be how several different RNA molecules catalyze the formation of bonds in the process of protein synthesis (translation) and/or mRNA processing. Specifically, the peptidyl transferase reaction is catalyzed by 23S rRNA, rather than by ribosomal proteins. Catalytic RNAs are also involved in splicing the introns of mRNA in *Archaea* and *Eukarya*.

Bloom's Taxonomy: 3-4: Applying/Analyzing  
Chapter Section: 4.10

14) You are trying to design a protein that will be expressed in *Escherichia coli* and secreted outside of the cell for purification and use as a pharmaceutical. *E. coli* is a gram-negative cell and the protein folds after it has exited the cell. Which secretion system would work best for this project? Support your answer with evidence based on the properties of *E. coli*, the protein, and the secretion system.

Answer: Answers should refer to the fact that *E. coli* is gram negative and thus has both a cytoplasmic and outer membrane. In order to secrete the protein a translocase that can move the protein through the outer membrane will be necessary. The protein folds after it has left the cell, thus the Sec system would work better than the Tat system, because the Tat system is specialized for proteins that fold in the cytoplasm and then cross the cytoplasmic membrane. Together these properties dictate that a two-step mechanism, such as a type II secretion system would work best for this project.

Bloom's Taxonomy: 5-6: Evaluating/Creating  
Chapter Section: 4.12



15) Explain how *Escherichia coli* can grow with a doubling time of 20 minutes when chromosome replication takes 40 minutes.

Answer: Answers should demonstrate understanding that a second round of replication will begin before the first round is finished under optimum growth conditions. Thus the cells can divide in less than 40 minutes, because it started copying the chromosome in the previous mother cell.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 4.3

16) Speculate why the half-life of mRNA is short, while the half-lives of rRNA and tRNA are long.

Answer: An important feature present in rRNA and tRNA yet lacking in mRNA is the secondary structures that make them more difficult to degrade by ribonucleases. These additional bonds also require more energy input to degrade them. Based on their function in the cell, mRNA is just a messenger and should thus be temporary. rRNA and tRNA are stable enzymatic molecules that are continuously catalyzing reactions in the cells.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 4.10

17) Speculate on why it can be problematic to copy and express genes from *Bacteria* into *Eukarya*.

Answer: Transcription in *Bacteria* is quite different than transcription in *Eukarya*. Different promoter sequences and RNA polymerases are used. In addition, mRNA in *Eukarya* must be tagged and modified so that it can be exported from the nucleus for subsequent translation. The promoter sequence and other signal sequences would have to be added to the *Bacterial* gene so that it would be properly transcribed. Another issue is that the universal genetic code does have exceptions. This means some mRNA could be translated into different proteins in different cells or not terminated at the correct location. Codon bias is another problem that can influence translational efficiency. The machinery and methods of post-translational modification are also very different and can therefore make cloning between *Bacteria* and *Eukarya* difficult.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 4.6

18) Explain why GC-rich DNA requires a higher temperature to denature or melt than AT-rich DNA and hypothesize as to why the GC content of chromosomes in microorganisms from different environments varies widely.

Answer: The overall increase in stability of G-C pairing should be emphasized over A-T binding. A DNA strand with high GC content (low AT) has more triple H bonds compared to an AT-rich (low GC) strand of the same length and contains mostly double H bonds. More energy is therefore required to break more H bonds, and thus GC-rich DNA is more heat resistant to denaturation. The fact that GC-rich DNA is more heat resistant may mean that microorganisms adapted to high temperature environments may have chromosomes with higher GC content than microorganism in moderate or low temperature environments.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 4.1